

BABOLSAR, Iran, Islamic Republic ofWMO: **407360**Lat: **36.6986N**Lon: **52.6425E**Elev: **-21**StdP: **101.58**Time Zone: **3.50 (IRN)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	3.1	-0.9	3.5	3.9	0.5	3.9	4.9	10.1	7.8	8.1	8.8	1.0	90	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	32.8	26.5	32.0	26.0	31.2	25.5	27.3	31.6	26.7	30.9	26.2	30.2	3.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.4	30.5	25.5	20.6	29.9	24.9	20.0	29.3	86.7	31.6	84.1	31.2	81.4	30.3	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.4	5.4	0.1	34.8	1.5	1.4	-1.0	35.9	-1.9	36.7	-2.7	37.5	-3.8	38.5		
(5)			DB	WB	-0.5	28.3	1.6	0.9	-1.6	29.0	-2.5	29.5	-3.4	30.0	-4.6	30.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	8.8	8.9	11.5	15.2	20.7	25.0	27.0	27.5	24.5	19.8	14.3	10.4		
	DBStd	7.27	2.47	2.46	2.87	2.88	2.60	1.92	1.64	1.61	2.24	2.73	3.00	2.46		
	HDD10.0	141	52	43	14	1	0	0	0	0	0	0	4	26		
	HDD18.3	1271	296	264	214	101	8	0	0	0	0	16	125	247		
	CDD10.0	3005	14	13	61	157	332	451	527	543	435	303	132	37		
	CDD18.3	1093	0	0	2	7	82	201	268	285	185	61	3	0		
	CDH23.3	9808	0	0	15	31	299	1643	2848	3252	1469	244	7	0		
	CDH26.7	3243	0	0	4	7	36	397	1044	1341	386	28	1	0		
(14) Wind	WSAvg	2.0	1.7	2.0	2.2	2.3	2.4	2.3	2.3	2.1	1.9	1.6	1.5	1.5		
(15) Precipitation	PrecAvg	791	81	60	59	35	24	29	30	30	91	116	141	96		
	PrecMax	1098	179	109	125	85	72	131	104	104	169	273	247	185		
	PrecMin	455	29	9	12	0	2	0	2	0	5	24	15	12		
	PrecStd	150	35	30	29	23	15	29	29	27	43	58	54	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.6	19.2	25.2	26.8	29.9	32.0	33.3	33.9	32.2	29.1	24.2	19.9		
		MCWB	12.1	12.9	16.2	18.6	22.3	25.4	26.7	27.2	25.9	22.6	19.7	14.4		
	2%	DB	15.5	15.8	20.6	23.1	27.3	30.9	32.6	33.0	31.0	27.0	21.8	17.1		
		MCWB	11.6	11.5	14.7	17.5	21.3	24.5	26.4	26.8	25.4	22.4	18.0	13.7		
	5%	DB	13.7	13.7	17.4	21.1	26.1	29.9	31.8	32.2	29.9	25.8	20.2	15.6		
		MCWB	10.9	10.6	13.4	16.8	21.0	24.0	25.8	26.3	24.7	21.6	16.9	12.9		
	10%	DB	12.5	12.3	15.1	19.3	25.0	29.0	30.9	31.7	28.9	24.6	19.0	14.3		
		MCWB	10.2	10.0	12.3	15.8	20.5	23.4	25.3	26.0	24.0	20.7	16.1	12.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	13.4	17.2	19.4	23.5	26.2	27.9	28.2	27.0	24.1	20.2	15.5		
		MCDB	16.3	18.0	23.7	24.8	27.5	30.5	32.3	32.5	31.1	26.9	23.0	18.0		
	2%	WB	12.2	11.9	14.9	18.1	22.4	25.4	27.1	27.6	26.0	22.9	18.5	14.2		
		MCDB	14.7	14.6	19.0	22.2	26.2	29.5	31.4	31.8	29.8	26.1	20.9	16.4		
	5%	WB	11.3	11.0	13.5	17.1	21.7	24.7	26.5	27.1	25.1	22.0	17.5	13.2		
		MCDB	13.4	13.1	17.1	20.4	25.2	28.7	30.6	31.3	29.0	25.0	19.8	15.1		
	10%	WB	10.5	10.4	12.5	16.1	20.9	24.1	25.9	26.5	24.5	21.2	16.4	12.4		
		MCDB	12.3	12.1	15.0	18.9	24.2	27.9	29.9	30.6	28.1	23.9	18.5	14.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	5.8	5.5	5.5	5.7	6.2	6.6	7.1	6.9	7.4	7.2	6.8		
		MCDBR	8.8	8.7	10.4	8.9	7.5	7.2	7.5	8.0	7.9	8.6	8.7	8.9		
	5% WB	MCWBR	5.8	5.5	5.7	4.4	3.7	3.2	3.3	3.9	4.0	4.9	5.5	6.1		
		MCWBR	8.0	7.4	9.5	8.0	6.8	6.6	7.0	7.6	7.4	8.0	7.9	7.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.401	0.445	0.502	0.575	0.581	0.589	0.637	0.601	0.576	0.530	0.455	0.408		
		taud	2.178	2.060	1.941	1.821	1.855	1.868	1.788	1.857	1.889	1.955	2.082	2.159		
	Ebn at Noon	Ebn at Noon	759	768	761	729	733	726	687	701	688	674	688	717		
		Edh at Noon	112	142	175	208	204	202	217	198	183	157	123	108		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.30	2.88	3.77	4.75	5.84	6.45	6.12	5.79	4.64	3.50	2.49	2.04		
		RadStd	0.20	0.32	0.32	0.53	0.32	0.40	0.49	0.62	0.29	0.26	0.21	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.29	N/A	-0.82	+0.62	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page